

Macroeconomics III

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Based on Chapter 3 of
your textbook

“CAPITAL
ACCUMULATION AND
GROWTH: THE
BASIC SOLOW
MODEL”

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THE BASIC SOLOW MODEL

- How can a nation become rich, i.e., initiate a growth process leading to higher GDP/consumption per capita in the long run?
 - The basic Solow model provides some first answers: It predicts how the evolution and the long-run levels of GDP and consumption per capita depend on structural parameters such as the rate of investment and the growth rate of the labour force.
 - Key elements of the Solow model:
 - In each period, output is determined by the supplies of capital and labour through the production function.
 - Exogenous savings/investment rate, s , exogenous growth rate of labour force, n , and exogenous depreciation rate, δ .
 - Explicit description of capital accumulation: $K_{t+1} = K_t + I_t - \delta K_t$
 - Accumulation of capital is the main driving force for wealth.
- “Basic” model: No technological progress

THE "MICRO WORLD" OF THE SOLOW MODEL

- **Object:** Closed economy
- **Time:** A sequence of periods/years $t = 0, 1, 2, \dots$
- **Agents:** Households and firms (and government)
- **Commodities and markets:** Output, capital services and labour services (one asset = physical capital)
- **The market for output:** Supply = firms' output, Y_t . Demand from households for consumption and investment = $C_t + I_t$. Relative price = 1.
 - One-sector model: Output can be used either for consumption or for investment.

- **The market for capital services:** Consumers own the capital stock, K_t , and rent its services to firms. Supply of capital services = K_t^s . Firms' demand = K_t^d
 - Relative price (in units of output) for renting one unit of capital for one period: r_t = real rental rate for capital
 - Real interest rate: $\rho_t = r_t - \delta$, where δ is the rate of depreciation, or: $r_t = \rho_t + \delta$
(Alternative interpretation: The firms own the capital, borrow for the purchase of capital at an interest rate of ρ_t and bear the cost of depreciation themselves.)
 - User cost = $r_t = \rho_t + \delta$
- **Labour market:** Households supply = L_t = the labour force. Demand from firms = L_t^d . Relative price: w_t = the real wage rate.
- **Competitive markets:** r_t and w_t adjust to equate supply and demand in all markets $\Rightarrow$ full (or natural) utilization of resources.

THE PRODUCTION SIDE

- is modelled as if all production (all of GDP) comes from one profit-maximizing firm that produces value added, Y_t , from capital services, K_t^d (machine-years), and labour services, L_t^d (man-years), according to the production function:

$$Y_t = F(K_t^d, L_t^d)$$

1. Constant returns to scale: $F(\lambda K_t^d, \lambda L_t^d) = \lambda F(K_t^d, L_t^d)$.
The replication argument!
2. Positive marginal products:

$$F'_K(K^d, L^d) > 0, \quad F'_L(K^d, L^d) > 0.$$

3. Marginal products are decreasing in the amount of the factor used:

$$F''_{KK} < 0, \quad F''_{LL} < 0$$

(diminishing returns) and growing in the amount of the other factor used:

$$F''_{LK} > 0, \quad F''_{LK} > 0$$

- Profit maximization: Given r_t and w_t , the firm chooses Y_t, K_t^d and L_t^d to:

$$\max Y_t - r_t K_t^d - w_t L_t^d, \quad \text{s.t. } Y_t = F(K_t^d, L_t^d)$$

The usual necessary conditions for an optimum:

$$F'_K(K_t^d, L_t^d) = r_t, \quad F'_L(K_t^d, L_t^d) = w_t$$

(These two equations do not determine K_t^d and L_t^d from given r_t and w_t ; *they only determine* K_t^d / L_t^d .)

- Competitive market clearing $\Rightarrow K_t^d = K_t$ and $L_t^d = L_t$, where K_t and L_t are the supplies in period t :

$$F'_K(K_t, L_t) = r_t, \quad F'_L(K_t, L_t) = w_t$$

Since K_t and L_t are predetermined in any given period, r_t and w_t are determined this way.

K_t and L_t predetermined. What does that mean?

THE INCOME DISTRIBUTION

$$r_t = F'_K(K_t, L_t) \Rightarrow r_t K_t = F'_K(K_t, L_t) K_t$$

$$w_t = F'_L(K_t, L_t) \Rightarrow w_t L_t = F'_L(K_t, L_t) L_t$$

1. No pure profits. Euler's rule $\Rightarrow$

$$F(K_t, L_t) - F'_K(K_t, L_t) K_t - F'_L(K_t, L_t) L_t = 0$$

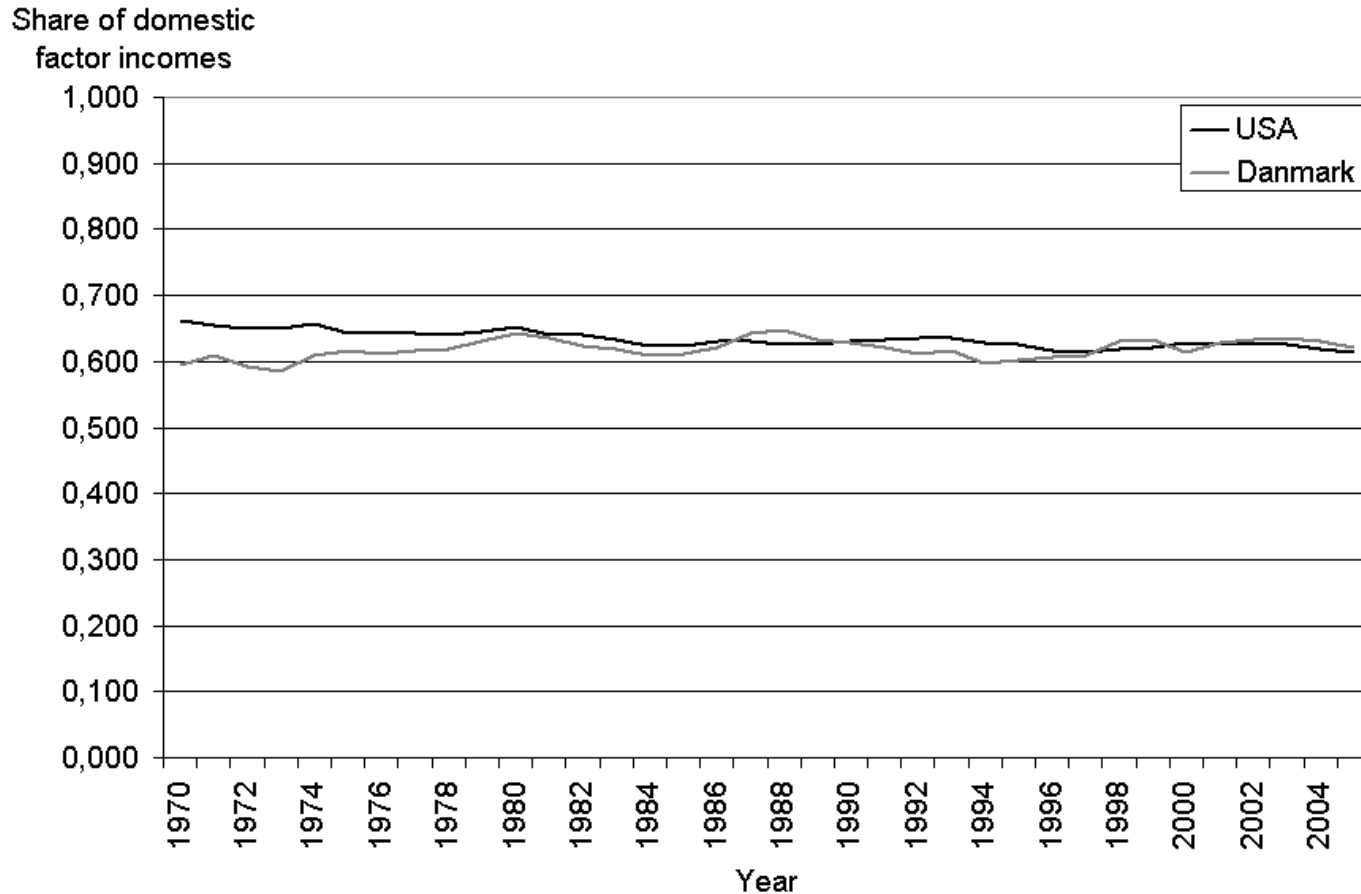
2. The functional income distribution

$$\frac{r_t K_t}{Y_t} = \frac{F'_K(K_t, L_t) K_t}{F(K_t, L_t)} \quad \text{and} \quad \frac{w_t L_t}{Y_t} = \frac{F'_L(K_t, L_t) L_t}{F(K_t, L_t)}$$

The income share of each factor is the elasticity of the production function with respect to the factor in question.

According to empirics, labour's share is relatively constant around $\frac{2}{3}$ over long periods except for short-run fluctuations:

Labour's share of domestic factor incomes



Is there a production function that fulfills all of our assumptions and has fixed output elasticities independently of K_t and L_t ? Yes, the Cobb-Douglas production function:

$$F(K_t, L_t) = B_t K_t^\alpha L_t^{1-\alpha}, \quad B_t > 0, \quad 0 < \alpha < 1,$$

where B_t is total-factor-productivity (TFP).

Check: It follows that

$$r_t = F'_K = \alpha B_t \left(\frac{K_t}{L_t} \right)^{\alpha-1} \quad \text{and} \quad w_t = F'_L = (1-\alpha) B_t \left(\frac{K_t}{L_t} \right)^{\alpha} \Rightarrow$$
$$\frac{r_t K_t}{Y_t} = \alpha \quad \text{and} \quad \frac{w_t L_t}{Y_t} = 1 - \alpha$$

The Cobb-Douglas function seems to be a realistic long-run assumption. We even have reason to believe that $\alpha \approx 1/3$.

HOUSEHOLDS

- The number of households in period t is L_t , which is predetermined. Household behaviour:
 1. Each supplies one unit of labour inelastically. Total supply $= L_t$.
 2. Owns the capital stock, K_t , which is predetermined in period t . Supply $= K_t$ (as long as $r_t > 0$).
 3. The representative household decides C_t given Y_t , and hence $S_t \equiv Y_t - C_t$. The **intertemporal budget constraint**:

$$K_{t+1} - K_t = S_t - \delta K_t, \quad 0 < \delta < 1$$

We **assume** that the result of the consumer's considerations is:

$$S_t = sY_t.$$

4. "Biology":

$$L_{t+1} = (1+n)L_t, \quad n > -1$$

THE COMPLETE MODEL (with book equation numbers below)

$$Y_t = BK_t^\alpha L_t^{1-\alpha} \quad (14)$$

$$r_t = \alpha B \left(\frac{K_t}{L_t} \right)^{\alpha-1} \quad (15)$$

$$w_t = (1-\alpha) B \left(\frac{K_t}{L_t} \right)^\alpha \quad (16)$$

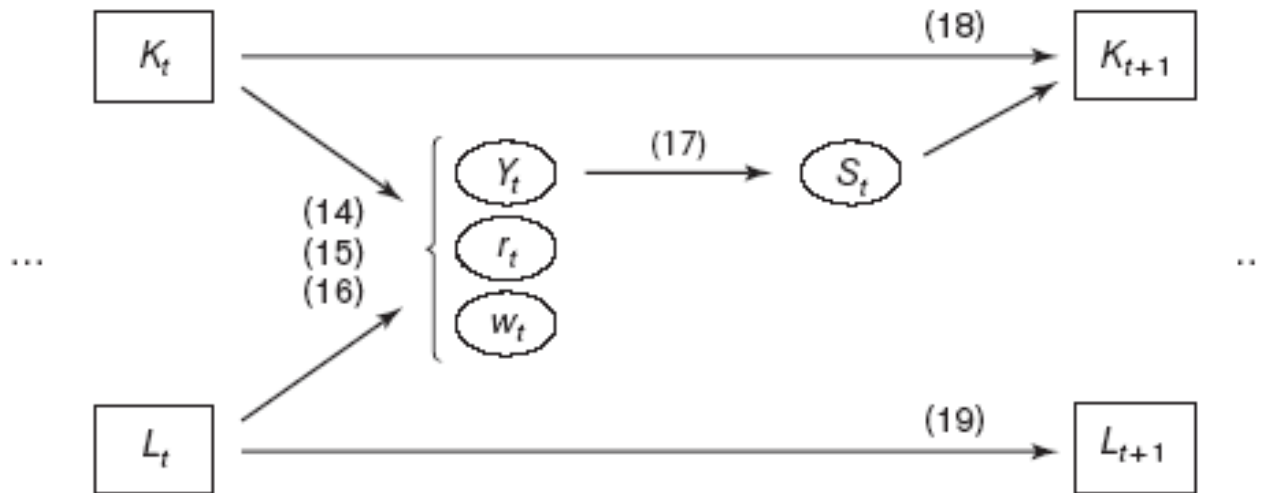
$$S_t = sY_t \quad (17)$$

$$K_{t+1} - K_t = S_t - \delta K_t \quad (18)$$

$$L_{t+1} = (1+n)L_t, \quad (19)$$

- Parameters: B, α, s, δ and n . N.b. No subscript t on B : "Basic" Solow model.
- Endogenous variables: $(Y_t), (K_t), (L_t), (r_t), (w_t)$ and (S_t) of which (K_t) and (L_t) are state variables:

- Given K_0 and L_0 the model determines $(Y_t), (K_t), (L_t) \dots$



- Government in the model? Yes. Simply interpret S_t as private plus government savings.
- We viewed the capital accumulation equation:

$$K_{t+1} - K_t = S_t - \delta K_t$$

as the household's budget constraint.

Alternatively: By definition we have that

$$K_{t+1} - K_t = I_t - \delta K_t \quad (*)$$

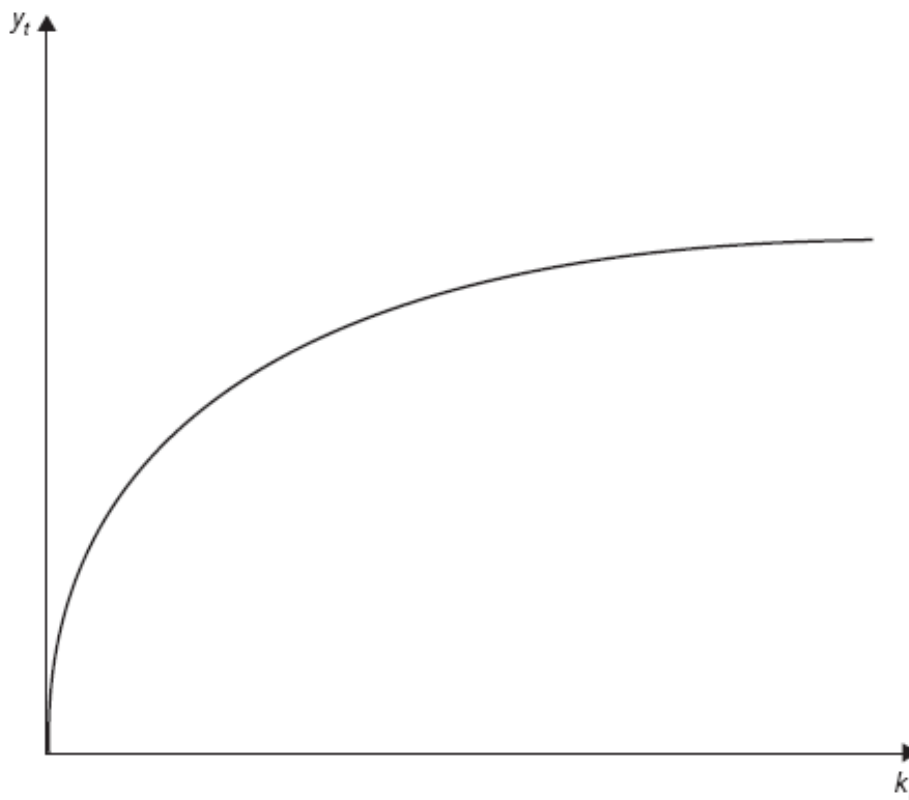
The condition for equilibrium in the output market, or the national accounting identity, is: $Y_t = C_t + I_t$, and by definition: $S_t \equiv Y_t - C_t$. Hence

$$I_t = S_t \quad (**)$$

Combining (*) and (**) gives the capital accumulation equation again: s is the savings rate and the investment rate.

ANALYSING THE BASIC SOLOW MODEL

1. Define: $y_t = Y_t / L_t$ and $k_t = K_t / L_t$.
2. From $Y_t = BK_t^\alpha L_t^{1-\alpha}$ we get the per capita production function: $y_t = Bk_t^\alpha$, $0 < \alpha < 1$



Note: $\ln y_t - \ln y_{t-1} = \alpha (\ln k_t - \ln k_{t-1}) \Leftrightarrow g_t^y = \alpha g_t^k$

3. Insert $S_t = sY_t$ into $K_{t+1} - K_t = S_t - \delta K_t$ to get:

$$K_{t+1} = sY_t + (1 - \delta)K_t$$

3. Divide by $L_{t+1} = (1+n)L_t$ on both sides to find that:

$$k_{t+1} = \frac{1}{1+n} (sy_t + (1-\delta)k_t)$$

4. Insert $y_t = Bk_t^\alpha$ to get the **transition equation**:

$$k_{t+1} = \frac{1}{1+n} (sBk_t^\alpha + (1-\delta)k_t)$$

5. Subtracting k_t from both sides of the transition equation gives the **Solow equation**:

$$k_{t+1} - k_t = \frac{1}{1+n} \left(sBk_t^\alpha - \underbrace{(n + \delta)k_t} \right)$$

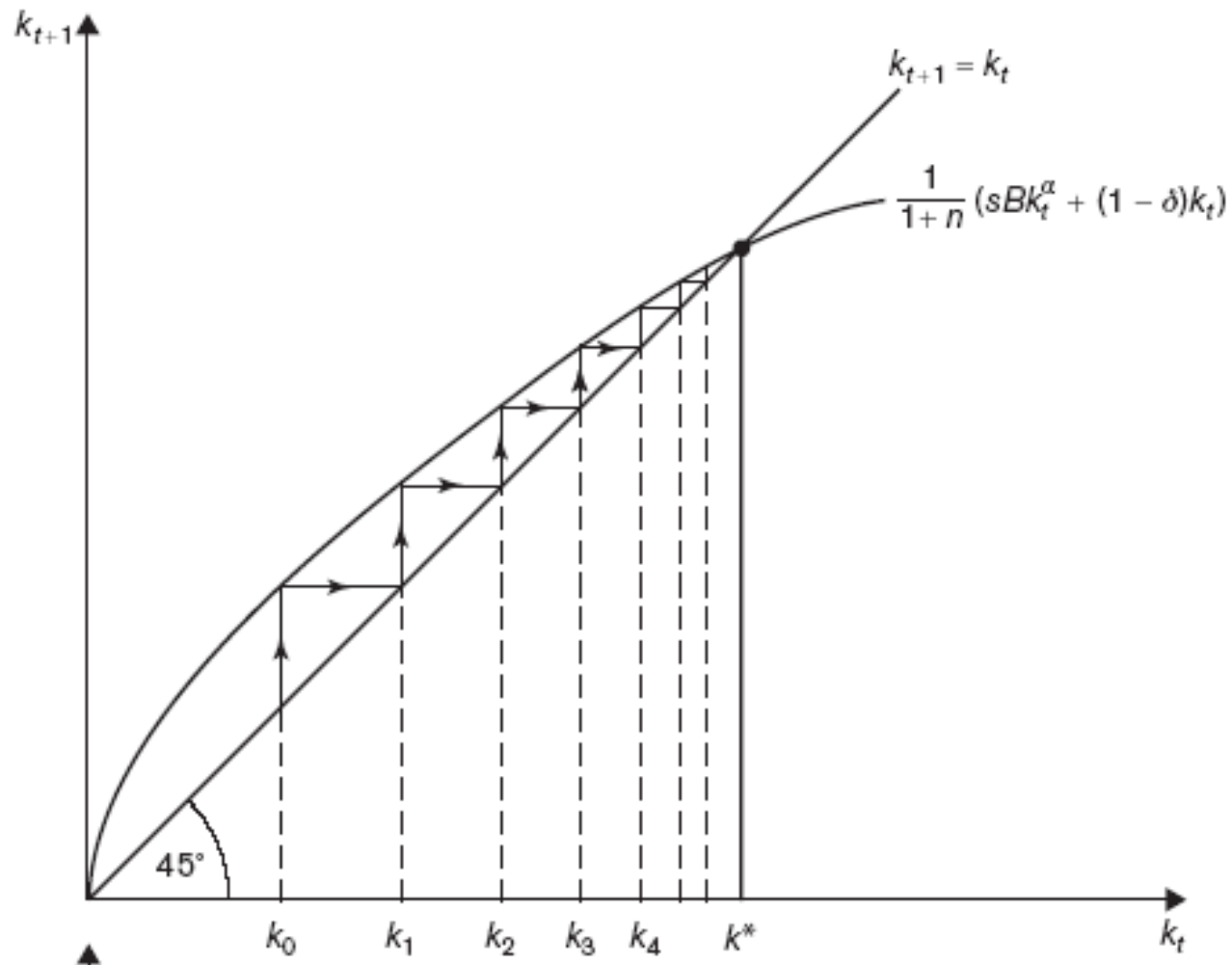
"technical term"
appearing because
of discrete time

savings per
capita = sy_t

Replacement investment to
compensate for depreciation
and growth of labour force

The transition diagram

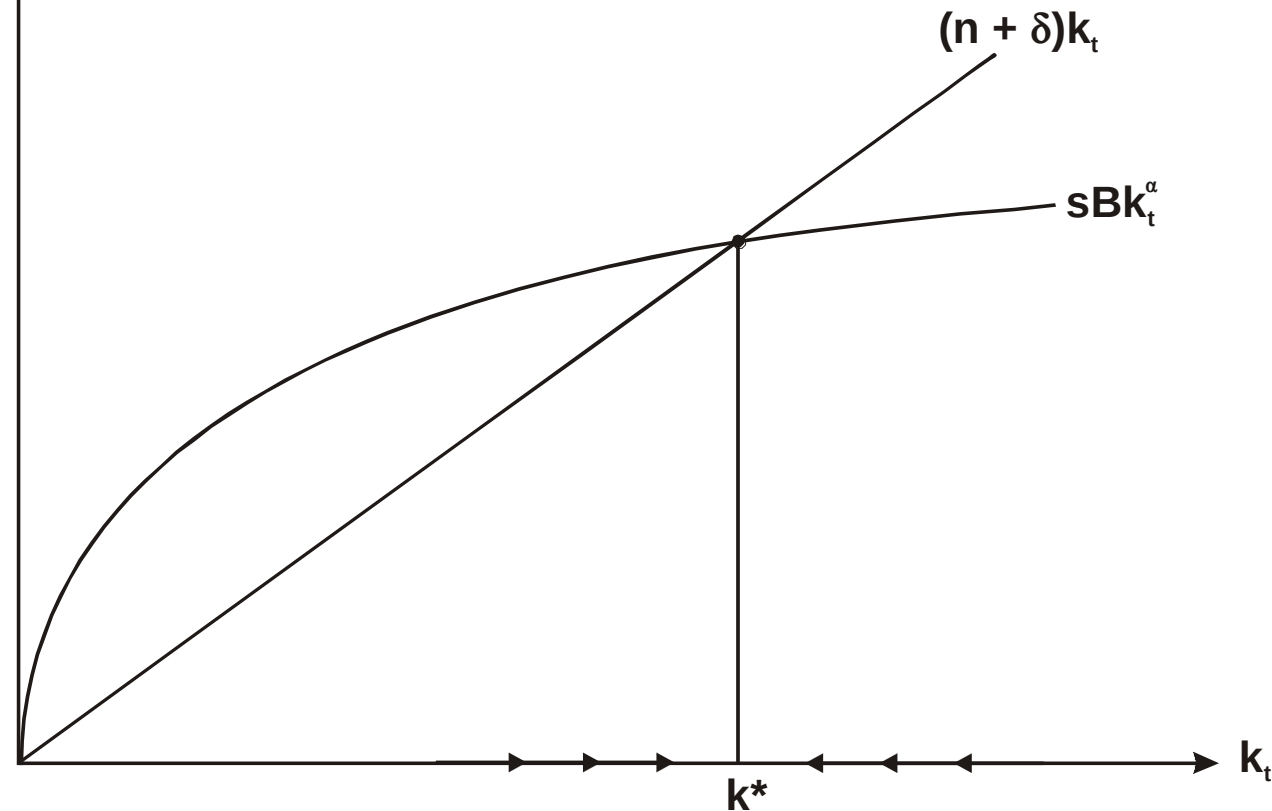
$$k_{t+1} = \frac{1}{1+n} (sBk_t^\alpha + (1-\delta)k_t)$$



- About the slope, dk_{t+1} / dk_t :
 - It is everywhere strictly positive
 - It goes to infinity as k_t goes to zero
 - It goes to $(1-\delta)/(1+n)$ as $k_t \rightarrow \infty$, and $(1-\delta)/(1+n) < 1 \Leftrightarrow n + \delta > 0$. Indeed we assume that $n > -\delta$ (realistic!).
- The figure shows that $k_t \rightarrow k^*$ in the long run. Hence $y_t \rightarrow y^* = B(k^*)^\alpha$, $c_t \rightarrow c^* = (1-s)y^*$ and $r_t \rightarrow r^* = \alpha B(k^*)^{\alpha-1}$ etc.
- The values k^*, y^* , etc. define the "steady state".

THE SOLOW DIAGRAM

$$k_{t+1} - k_t = \frac{1}{1+n} (sBk_t^\alpha - (n+\delta)k_t)$$



Why does growth in k_t and y_t have to stop?
Diminishing returns!

STEADY STATE

- The long run levels k^* , y^* , etc. depend on parameters. How? What makes a nation rich?
- Look at the Solow equation:

$$k_{t+1} - k_t = \frac{1}{1+n} \left(sBk_t^\alpha - (n + \delta)k_t \right)$$

In steady state $k_{t+1} - k_t = 0$. Insert this to find

$$sB(k^*)^\alpha = (n + \delta)k^* \Leftrightarrow$$

$$k^* = B^{\frac{1}{1-\alpha}} \left(\frac{s}{n + \delta} \right)^{\frac{1}{1-\alpha}} \Rightarrow$$

$$y^* = B(k^*)^\alpha = B^{\frac{1}{1-\alpha}} \left(\frac{s}{n + \delta} \right)^{\frac{\alpha}{1-\alpha}}$$

- Some sharp predictions of the Solow model:

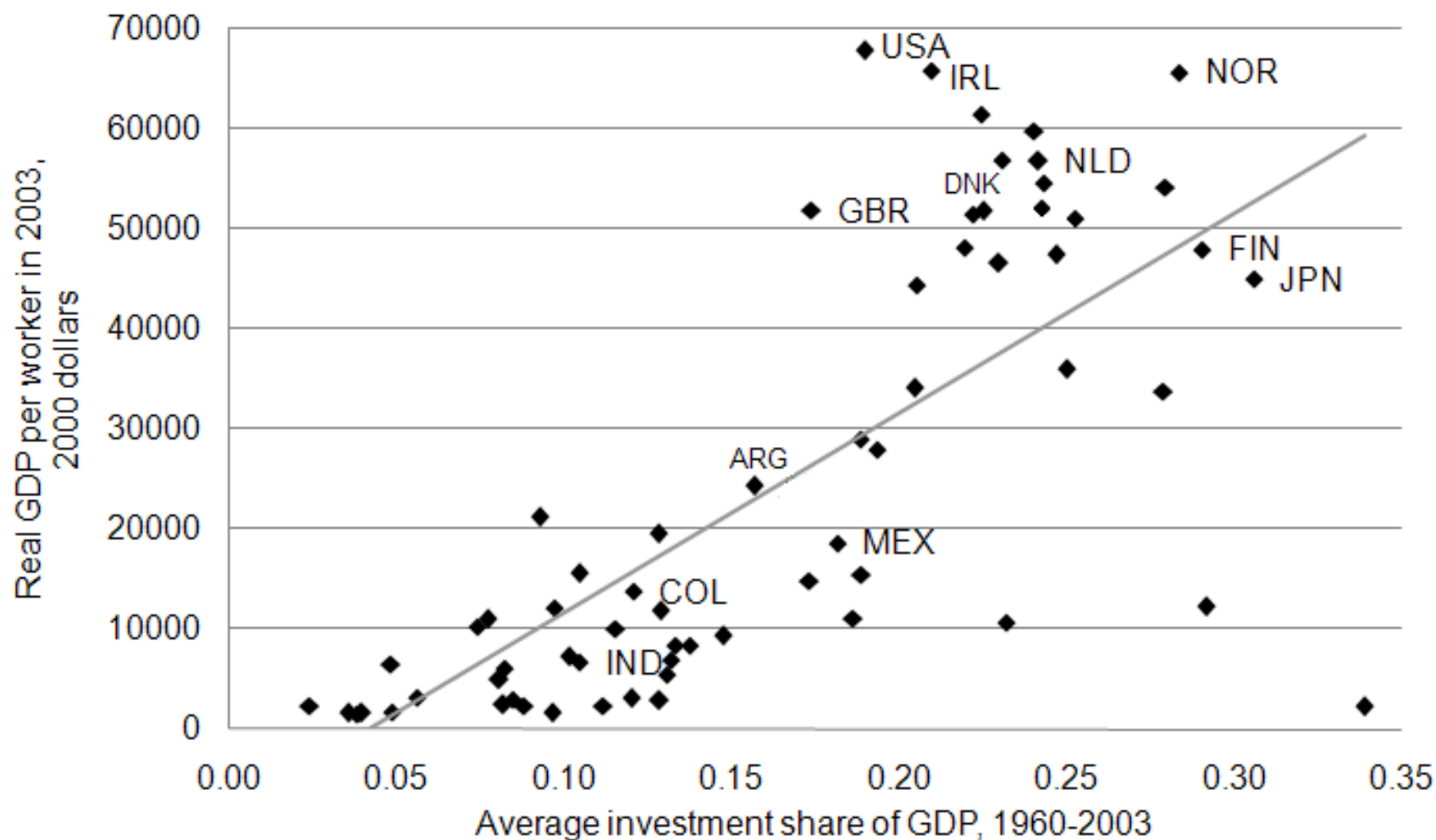
$$\ln y^* = \frac{1}{1-\alpha} B + \frac{\alpha}{1-\alpha} [\ln s - \ln(n + \delta)].$$

The elasticity of y^* wrt. s is $\frac{\alpha}{1-\alpha} \approx \frac{1}{2}$ (since we believe that $\alpha \approx \frac{1}{3}$): an increase in s of 10%, e.g., from 20 to 22%, should give an increase in y^* of 5%!

The elasticity of y^* wrt. $n + \delta$ and wrt. B are $-\frac{\alpha}{1-\alpha} \approx -\frac{1}{2}$ and $\frac{1}{1-\alpha} \approx \frac{3}{2}$, respectively. Why is the latter not one? Capital accumulation.

- We have reached empirically testable hypotheses!
Empirics:

Real GDP per worker against the average investment share across 65 countries



Real GDP per worker against the average annual labour force growth rate across 65 countries

